Econ 101A
Section 11

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Last time we stopped at the cost function. Today we finish the firm. We pick output, read the supply curve off the cost curves, and then solve the whole thing in one step to check that we get the same answer. One theme runs through everything: returns to scale decide whether the firm's problem has a nice interior answer. Under perfect competition, only decreasing returns give a well-behaved supply curve.

1 Cost Minimization, Step 2: Choosing Output

Given the cost function $c(w, r, y)$ and an output price p the firm takes as given, the firm solves

$$\max_{y \geq 0} \; py - c(w, r, y).$$

First-order condition. $p - c'_y(w, r, y) = 0$, that is, **price equals marginal cost**.

Second-order condition. $-c''_{yy}(w, r, y^*) < 0$, that is, marginal cost must be *increasing* at y^*. If marginal cost is falling, the first-order condition finds a profit minimum.

Supply slopes up. The supply function $y^*(w, r, p)$ is defined by $F(y, p) \equiv p - c'_y(y) = 0$. By the implicit function theorem, with $F_p = 1$ and $F_y = -c''_{yy}$,

$$\frac{\partial y^*}{\partial p} = -\frac{F_p}{F_y} = \frac{1}{c''_{yy}} > 0 \quad \text{whenever the SOC holds.}$$

Returns to scale shape costs. For $f = AK^\alpha L^\beta$, Lecture 8 shows that

$$c(w, r, y) = B\Big(\frac{y}{A}\Big)^{\frac{1}{\alpha+\beta}}, \qquad B \equiv w\Big(\frac{w\alpha}{r\beta}\Big)^{-\frac{\alpha}{\alpha+\beta}} + r\Big(\frac{w\alpha}{r\beta}\Big)^{\frac{\beta}{\alpha+\beta}}.$$

The exponent $1/(\alpha + \beta)$ does all the work:

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- DRS ($\alpha + \beta < 1$): exponent above 1, cost is convex, MC rises. Interior optimum, supply well defined.
- CRS ($\alpha + \beta = 1$): cost is linear in y, MC is flat at B/A. Produce nothing if $p < B/A$, anything if $p = B/A$, "infinitely much" if $p > B/A$.
- IRS ($\alpha + \beta > 1$): cost is concave, MC falls. The FOC gives a minimum; profit grows without bound in y.

2 Cost Curves and the Supply Function

Marginal cost $\text{MC} = \partial c/\partial y$ tells you how much to produce: $p = \text{MC}$.

Average cost $\text{AC} = c/y$ tells you whether to produce at all. Dividing profit by $y > 0$,

$$\pi = py - c > 0 \iff p - \frac{c}{y} > 0 \iff \text{AC} < p.$$

MC cuts AC at its minimum. Differentiate AC with the quotient rule:

$$\frac{d}{dy}\Big(\frac{c(y)}{y}\Big) = \frac{c'(y)\,y - c(y)}{y^2} = \frac{1}{y}\Big(c'(y) - \frac{c(y)}{y}\Big) = \frac{\text{MC} - \text{AC}}{y}.$$

AC falls when MC is below it and rises when MC is above it. So the two curves cross where AC is flat, at its minimum.

The supply curve is the part of the MC curve that lies above AC. Below minimum AC, the firm shuts down and supplies zero.

3 One-Step Profit Maximization

Instead of two steps, choose inputs directly:

$$\max_{L,K}\ pf(L,K) - wL - rK.$$

First-order conditions. $pf'_L = w$ and $pf'_K = r$. Each input is hired until the *value of its marginal product* equals its price. Dividing one by the other gives back $f'_L/f'_K = w/r$: the one-step firm also minimizes cost.

Second-order conditions. $pf''_{LL} < 0$ and

$$|H| = \begin{vmatrix} pf''_{LL} & pf''_{LK} \\ pf''_{LK} & pf''_{KK} \end{vmatrix} = p^2\Big[f''_{LL}f''_{KK} - \big(f''_{LK}\big)^2\Big] > 0.$$

Diminishing marginal products are not enough. The cross effect f''_{LK} must not be too large.

Comparative statics. Differentiate the two FOCs with respect to w and to r, and solve with Cramer's rule:

$$\frac{\partial L^*}{\partial w} = \frac{p f''_{KK}}{|H|} < 0, \qquad \frac{\partial L^*}{\partial r} = \frac{-p f''_{LK}}{|H|}.$$

Labor demand always falls with its own price. The effect of the price of capital depends on the sign of f''_{LK}: if capital makes workers more productive ($f''_{LK} > 0$), a higher r lowers labor demand too.

4 Exercises

Exercise 1 (a well-behaved firm) Let $y = K^{1/4}L^{1/4}$, input prices $w, r > 0$, output price $p > 0$.

(a) What are the returns to scale? Find the conditional factor demands and the cost function.

(b) Find MC and AC. Which is larger?

(c) Solve for the supply function $y^*(w, r, p)$. Check the SOC and verify $\partial y^*/\partial p = 1/c''_{yy}$.

(d) Let $w = r = 1$ and $p = 8$. Find y^*, L^*, K^* and profit.

Exercise 2 (what goes wrong with CRS and IRS)

(a) Let $y = K^{1/2}L^{1/2}$. In Section 10 we found $c = 2y\sqrt{wr}$. Find MC and AC, and solve the output choice for every p.

(b) Let $y = KL$. Find the conditional factor demands and the cost function. Find the y that solves the FOC. Is it a maximum? What is profit there? What does the firm do?

Exercise 3 (cost curves with one input and a setup cost) One input, labor, with wage w. Technology $y = L^\alpha$, $\alpha > 0$.

(a) Find $c(w, y)$, MC and AC. Show that MC $=$ AC$/\alpha$.

(b) Let $\alpha = \frac{1}{2}$ and $w = 2$. Find the supply function.

(c) Describe MC, AC and supply when $\alpha = 1$ and when $\alpha > 1$.

(d) Keep $\alpha = \frac{1}{2}$, $w = 2$, but producing any positive output requires a setup cost $F = 8$, which the firm avoids if it produces nothing. Find MC, AC, the minimum of AC, and the supply function. Sketch the curves.

Exercise 4 (one step, same answer) Take the firm of Exercise 1, $y = K^{1/4}L^{1/4}$, and solve $\max_{L,K}\ pK^{1/4}L^{1/4} - wL - rK$ directly.

(a) Write the FOCs and solve for $L^*(w, r, p)$ and $K^*(w, r, p)$. Check against Exercise 1(d).

(b) For general $f = AK^\alpha L^\beta$, show that $f''_{LL}f''_{KK} - (f''_{LK})^2 = \alpha\beta(1 - \alpha - \beta)A^2K^{2\alpha-2}L^{2\beta-2}$. When do the SOCs hold?

(c) Sign $\partial L^*/\partial w$ and $\partial L^*/\partial r$. Compare with the conditional demand $L^*(w, r, y) = y^2(r/w)^{1/2}$ from Exercise 1.

(d) At $p = 8$, $w = r = 1$, verify that $\left.\frac{\partial L^*}{\partial r}\right|_{\text{one-step}} = \left.\frac{\partial L^*}{\partial r}\right|_{y \text{ fixed}} + \left.\frac{\partial L^*}{\partial y}\right|_{\text{cond.}} \cdot \frac{\partial y^*}{\partial r}$. Interpret.